Venkata Lakshmi Parimala Pasupuleti

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PROFESSIONAL SUMMARY

AI engineer with 9 years of production software engineering (Python, TypeScript, SQL) and hands-on agentic AI development - with a focus on prompt engineering, agent behavior tuning, and quality measurement. Built LeaseI, an 8-agent LangGraph pipeline where each agent's prompts, guardrails, and conditional routing were hand-engineered and evaluated against a pass@k harness with LLM-as-judge scoring (100% citation correctness, 0% unsupported-claim rate). Built AgentProbe, a CI-integrated 5-metric eval framework that blocks regressions before merge. At Entrata, owned cross-functional delivery of 20+ production modules for 100K+ MAU - translating messy stakeholder feedback into scoped, shippable product improvements across 120+ Agile sprints, including Salesforce CRM integrations and workflow automation with Clay and n8n. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

CORE COMPETENCIES

Prompt Engineering | Agent Behavior Tuning | LLM Workflows & Guardrails | HITL Escalation Design | Eval / Rubric / LLM-as-Judge | Agent Quality Measurement | Multi-Agent Orchestration | Customer Feedback Triage | LangGraph / LangChain | FastAPI / REST APIs | Python / TypeScript / SQL | CRM & Workflow Automation | Salesforce / Clay / n8n | Product Judgment

EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer Intern

- **Prompt-engineered and tuned each stage of a LangGraph agentic pipeline** (parse → plan → retrieve → validate → conditional routing) - writing, testing, and iterating on prompts, instructions, and guardrails until agent behavior matched target outcomes across diverse real-world inputs.
- Designed and enforced behavioral guardrails in code - FAA altitude limits, battery budgets, geometric no-fly-zone checks - blocking non-compliant agent outputs before export; every compliance check structured as a grounded hypothesis test against a cited regulation corpus.
- Built a RAG layer (ChromaDB + FAA Part 107 corpus) with prompt-level controls ensuring every agent answer was cited to an exact source or explicitly flagged as unsupported - zero hallucination tolerance in regulated output paths.

Entrata - PropTech / Real Estate SaaS

Jan 2017 - Aug 2024

Senior Software Engineer (promoted from Trainee → Associate → Senior)

- **Translated messy stakeholder feedback into clear, shippable improvements** - owned end-to-end delivery of 20+ production modules across 120+ Agile sprints for 100K+ MAU; triaged customer asks into what could ship now, what needed product changes, and what should wait.
- **Owned Salesforce CRM integrations** - built data pipelines connecting product data to commercial systems; configured custom objects and automated workflows using Clay and n8n, syncing operational data across teams without creating one-off complexity.
- **Shipped Entrata's first production AI pipeline** across 1.6M real financial records (Python, PyTorch, OpenAI API) - automated data processing workflows replacing manual Excel operations; built output validation, quality gates, and compliance controls to ensure agent behavior met production standards at launch.
- **Managed end-to-end compliance workflows** for affordable housing: HAP allocation, AMI-based pricing, HCV/Section 8/LIHTC certifications, HUD payments - translating federal and state regulatory requirements into scoped, testable product configurations; zero compliance violations.
- Optimized 200+ SQL queries (30% performance gain); led Docker microservices redesign (50% scalability improvement, 30% deployment time reduction); eliminated 80% of manual operational workload through Python/Bash automation; conducted 300+ peer code reviews.

George Mason University

Aug 2024 - May 2026

Research Associate & Graduate Teaching Assistant

- Built multi-source regression models ($R^2 = 0.977$ across 190+ countries) and applied Isolation Forest anomaly detection across 3 research pipelines - surfacing issues manual review missed; designed Tableau dashboards improving research comprehension 40% for faculty stakeholders.
- Mentored 600+ students across Python programming and database design - translating complex technical concepts into clear, actionable feedback for diverse learners.

PROJECTS

Leasel - AI Document Intelligence & 8-Agent Investigation Platform 2026

Python · LangGraph (8-agent StateGraph) · FastAPI · TF-IDF RAG · SQLite · OCR · Streamlit

Built a three-surface enterprise AI platform for affordable housing/LIHTC compliance. **Phase 1 - Document Q&A:** prompt-engineered a linear pipeline with section-aware chunking, TF-IDF retrieval, and strict output controls - 100% citation correctness, 100% grounding, 0% unsupported-claim rate on eval set. **Phase 3 - 8-agent LangGraph investigation graph:** each agent (Triage → Retrieval → Configuration → Historical Incident → Structured Data → Reasoning → Validation → Human Escalation) has individually tuned prompts, guardrails, and conditional routing; HITL escalation fires when confidence drops below threshold; validated end-to-end on a misconfigured-setting root-cause scenario. **Tenant Self-Service:** safety boundary tested - agent never auto-resolves eligibility or configuration questions for a tenant, even at high confidence. Core rule enforced in code: LLM reasons, human or deterministic logic makes every final regulated call.
github.com/venkatalakshmiparimala/Lease_complaince_rag

AgentProbe - Agentic AI Eval & Testing Toolkit Live

Python · LangGraph · FastAPI · ChromaDB · LLM-as-Judge · GitHub Actions · React

Built a CI-integrated eval framework to measure whether AI agents are actually getting better - the same problem Pam faces with voice agent quality. **5-metric LLM-as-judge pipeline** (factual accuracy, hallucination, retrieval, completeness, compliance) runs on every commit; regressions are blocked before merge. **3-agent LangGraph runtime** (Orchestrator, RAG Tracer, Auto Fix) with multi-provider routing (Anthropic, OpenAI, Groq, Ollama) and 15-endpoint FastAPI backend. React dashboard surfaces per-run metrics, regression drift, and retrieval quality - trace-level observability across concurrent agent test suites.
github.com/venkatalakshmiparimala/AgentProbe

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026

GPA: 3.87

B.Tech Computer Science & Engineering - Jawaharlal Nehru Technological University — 2017

TECHNICAL SKILLS

AI & Agents: Prompt Engineering, LangGraph, LangChain, Multi-Agent Orchestration, LLM Workflows, Agent Behavior Tuning, Guardrails, HITL Design, LLM-as-Judge, Eval Frameworks, RAG Pipelines, ChromaDB, Embeddings, NLP, OpenAI API, Anthropic API, Groq, Ollama

Languages: Python, TypeScript, JavaScript, SQL, PL/SQL, PHP, R, Bash

Backend & APIs: FastAPI, REST APIs, RabbitMQ, PostgreSQL, MySQL, SQLite, Oracle

CRM & Automation: Salesforce (integrations, custom objects, reporting automations, data sync), Clay, n8n, Zapier, CRM data pipelines, workflow automation

Cloud & Infra: AWS (S3, EC2, RDS, Glue), Docker, GitHub Actions (CI/CD), Microservices, ETL

ML & Data: scikit-learn, PyTorch, XGBoost, Isolation Forest, Random Forest, Feature Engineering, Pandas, NumPy, Tableau, Power BI